MILITARY SPECIFICATION

CASE, CARTRIDGE, 90MM, M19

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 This specification covers Case, Cartridge, 90MM, M19.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-A-2550 — Ammunition and Special Weapons, General Specification for.

MIL-P-10025 — Packing and Marking for Domestic Shipment of Inert Ammunition Components, General Specification for.

MIL-I-45208 — Inspection Requirements, General Specification for.

STANDARDS

FEDERAL

Federal Test Method Standard No. 151 — Metals; Test Methods.

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-109 — Quality Assurance Terms and Definitions.

MIL-STD-639 — Surface Roughness Values Preferred for Army Ordnance Materiel.

MIL-STD-1235 — Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes.

DRAWINGS

MUNITIONS COMMAND

F-7548413 — Case Cartridge, 90MM, M19.

F-7548320 — Cartons, Ammo. For Interplant Shipment of Brass, Aluminum and Spirally Wrapped Steel Cartridge Cases, Empty 75MM to 120MM Inclusive.

FSC 1315

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IEL–7548413— Inspection Equipment List for Case, Cartridge, 90MM, M19.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 General. The cartridge case shall comply with the requirements of the applicable drawings, referenced specifications and the following:

3.2 Initial production sample. Unless otherwise directed by the contracting officer, an initial production sample shall be required (see 6.1).

3.3 Material. The contractor shall be responsible for selecting from the classification list contained in the raw material specifications, the surface and physical condition of the raw material intended for use in manufacture.

3.4 Annealing. Annealing temperature shall not exceed 1200° Fahrenheit (F).

3.5 Stress relief. After the last cold forming operation, the cartridge case shall be given a stress relief treatment of approximately 500° F. for one hour.

3.6 Mouth anneal. After the final forming operation, the cartridge case shall be annealed at the mouth for a distance of 4 inches, minimum, 4½ inches, maximum.

3.7 Mercurous nitrate. The cartridge case shall show no evidence of cracking after immersion in mercurous nitrate.

3.8 Maximum profile. The maximum profile of the cartridge case shall comply with the requirements of the applicable drawings.

3.9 Mechanical properties.

3.9.1 *Hardness.* The metal .120 inch minimum, from the surface around the primer hole shall have a Rockwell Hardness Value of not less than B65.

3.9.2 *Tensile strength.* Specimens taken from sample cartridge cases in accordance with fig. 1 shall comply with the requirements of table I (see 6.2).

TABLE I

Specimen	Pounds per square inch (PSI)
A	44,000 Minimum—58,000 Maximum
B	50,000 Minimum
C	65,000 Minimum
D	65,000 Minimum
	57,000 Minimum

Elongation shall be recorded for information only.

3.10 Functional. The cartridge case shall be subjected to a ballistic test in the following phases:

(a) Excess pressure phase.

(b) Excess, excess pressure phase (for information only).

3.10.1 *Obturation.* The cartridge case shall effectively seal the breech of the weapon against gas generated during firing.

3.10.2 *Soundness.* After firing, the cartridge case shall be free of splits or cracks.

3.10.3 *Ejection.* When fired, the cartridge case shall be ejected by the service ejector when the weapon is operated in the normal manner (see 6.3).

3.10.4 *Primer retention.* The cartridge case shall retain the primer during firing and ejection.

3.11 Workmanship. The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

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3.11.1 *Metal defects.* The cartridge case shall be free from cracks, blisters, seams, laminations, deep scratches and porosity.

3.11.2 *Folds and cold shuts.* The cartridge case shall contain no folds or cold shuts in the radius area extending beyond the tangents from the base and sidewall. However, at no time will any fold or cold shut exceed .035 inches, maximum length (fig. 2).

3.11.3 *Bur.* No cartridge case shall have a bur which might interfere with the assembly of the primer or projectile, or which might be injurious to personnel handling the cartridge case.

3.11.4 *Foreign matter.* No cartridge case shall contain chips, dirt, grease, oil, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 *Quality assurance terms and definitions.* Reference shall be made to MIL–STD–109 to define quality assurance terms used.

4.1.2 *Inspection.* Inspection shall be in accordance with MIL–STD–2550 and MIL–I–45208, except as specified herein.

4.2 Initial production sample. Prior to initiation of regular production, the contractor shall submit an initial production sample to a Government approved facility for evaluation. The sample shall consist of seventy cartridge cases for performance of examination and tests as specified in 4.3 utilizing the procedures of 4.4 except as provided in MIL–I–45208. The Government will perform the functional tests on sixty cartridge cases in accordance with 4.3.4.3 utilizing the procedures in 4.4.5 to determine compliance with 3.10. The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as will be used in regular production. All parts and materials, including packaging and packing, must be the same as will be used for regular production and shall be obtained from the same source of supply. The contractor shall inspect the sample in accordance with his inspection system and other inspection requirements of MIL–I–45208. After inspection and provisional acceptance at source, the initial production sample shall be inspected at the Government approved facility for all requirements of the drawings and specifications. The Government reserves the right to require new initial production samples until such time as an acceptable sample is submitted.

4.3 Inspection provisions.

4.3.1 *Presentation of lots.* The product shall be presented in accordance with MIL–STD–105 or MIL–STD–1235, as applicable.

4.3.2 *Examination.* One hundred percent examination shall be performed for all critical defects. Examination for Major and Minor Defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plans contained in MIL–STD–105 or MIL–STD–1235, as applicable. An AQL of 0.25 percent for each Major Defect and an AQL of 0.40 percent for each Minor Defect shall be used. All nonconforming material shall be rejected. Four power (4X) magnification may be used in performing visual examination for such defects as bur and foreign matter. The Ordnance Standard Finish Blocks prescribed in MIL–STD–639 shall be used as a basis of comparison for surface roughness determination.

4.3.2.1 *Classification of defects.*

4.3.2.1.1 *Case, cartridge, drawing F–7548–413.*

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Defect	Method of inspection
Critical: None defined.	
Major:	
101. Primer hole not functionally concentric with body.	Gage
102. Primer hole counterbore not functionally concentric with primer hole.	Gage
103. Diameter of flange, min.	Gage
104. Diameter of shoulder under flange, min.	Gage
105. Diameter of mouth	Gage
106. Diameter of primer hole	Gage
107. Depth of primer hole counterbore, min.	Gage
108. Distance from bottom of primer hole counterbore to forward surface of flange, min.	Gage
109. Length of shoulder in primer hole counterbore, min.	Gage
110. Taper in primer hole, incorrect	Gage
111. Thickness of wall at mouth	Gage
112. Thickness of wall at rear length datum	Gage
113. Thickness of wall at forward length datum.	Gage
114. Thickness of flange, min.	Gage
115. Thickness of flange and shoulder, min.	Gage
116. Thickness through head	Gage
117. Interior contour at juncture of head and sidewall not smooth.	Visual
118. Metal defective	Visual
Minor:	
201. Diameter of primer hole counterbore.	Gage
202. Diameter of body under shoulder, min.	Gage
203. Length from head first datum diameter.	Gage
204. Length from head second datum diameter.	Gage
205. Length from head third datum diameter.	Gage
206. Length from head fourth datum diameter.	Gage
207. Total length	Gage
208. Radii or chamfers, missing or incorrect.	Visual
209. Surface finish, improper	Visual/ Measure
210. Dent or damaged	Visual
211. Marking missing, incorrect or illegible.	Visual
212. Bur, except as permitted	Visual
213. Foreign matter	Visual

4.3.2.1.2 *Carton, drawing F–7548320 — prior to sealing.*

Defect	Method of inspection
Critical: None defined.	
Major: None defined.	
Minor:	
201. Improper assembly	Visual
202. Support missing or damaged	Visual
203. Separator missing or damaged	Visual
204. Incorrect number in carton	Visual

4.3.2.1.3 *Carton, drawing F–7548320 — after sealing.*

Defect	Method of inspection
Critical: None defined.	
Major: None defined.	
Minor:	
201. Carton damaged, cut or gouged through all layers.	Visual
202. Carton not completely sealed	Visual
203. Contents loose	Manual/ Visual
204. Marking missing, incorrect or illegible.	Visual

4.3.3 *Maximum profile.* Each cartridge case shall be gaged in a maximum profile gage. Failure of any cartridge case to freely enter the gage in such a manner as to permit the slide to close without interference, shall be cause for rejection of the cartridge case.

4.3.4 *Inspection testing.* The following tests shall be performed in accordance with 4.4, except as provided in MIL–I–45208.

4.3.4.1 *Mercurous nitrate.* Five cartridge cases from the initial production sample and five cartridge cases from each production lot shall be subjected to mercurous nitrate immersion. Failure of one or more cartridge cases to comply with the requirements of 3.7 shall be cause for rejection of the initial production sample or the production lot.

4.3.4.2 *Mechanical properties.*

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4.3.4.2.1 *Hardness.* Five cartridge cases from the initial production sample and five cartridge cases from each production lot shall be subjected to this test. Failure of one or more cartridge cases to comply with the requirements of 3.9.1 shall be cause for rejection of the initial production sample or the production lot.

4.3.4.2.2 *Tensile strength.* Five cartridge cases from the initial production sample and five cartridge cases from each production lot shall be subjected to this test. Failure of one or more cartridge cases to comply with the requirements of 3.9.2 shall be cause for rejection of the initial production sample or the production lot. The cartridge cases used for the hardness test may be used for this test.

4.3.4.2.3 *Elongation.* The percentage of elongation of each specimen used for the tensile test shall be measured and recorded for information only.

4.3.4.2.4 *Macro examination.* Five cartridge cases from the initial production sample and five cartridge cases from each production lot shall be subjected to this test. Failure of one or more cartridge cases to comply with the requirements of 3.11.2 shall be cause for rejection of the initial production sample or the production lot. The cartridge cases used for the hardness test may be used for this test.

4.3.4.3 *Functional.* The following tests will be performed by the Government.

4.3.4.3.1 *Initial production sample.* The quantities listed in table II will be ballistically tested in the phases indicated and the results of the test will be evaluated in accordance with the acceptance criteria set forth in table III, except that no retest will be permitted.

TABLE II

Test phase	Quantity to be tested
Excess pressure phase	30
Excess, excess pressure phase (for information only).	30

4.3.4.3.2 *Production lot sample.* Thirty cartridge cases from each production lot of 25,000 or less shall be ballistically tested in the excess pressure phase. For lots which exceed 25,000 cartridge cases, one additional cartridge case will be tested for every one thousand cartridge cases (or fraction thereof) above 25,000. Acceptance and rejection shall be in accordance with table III. The lot shall be considered eligible for retest, using double the number of samples, provided that after firing, no cartridge case contains crack(s) or split(s) within one caliber of the head, and the rejection criteria of table III has not been attained.

4.3.5 *Process control.* Monitoring at unscheduled intervals shall be conducted for the following process operations to determine compliance with the following requirements:

Anneal — 3.4.
Stress relief — 3.5.
Mouth Anneal — 3.6.

TABLE III

Defect characteristic	First sample		Cumulative sample	
	Accept	Reject	Accept	Reject
1. Obturation	0	2	1	2
2. Splits or cracks within 1 caliber of head.	0	1	Not permitted	Not permitted.
3. Splits or cracks in sidewall more than 1 caliber from head.	1	3	2	3
4. Ejection	0	2	1	2
5. Primer retention	0	2	1	2

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Failure to comply with requirements shall be cause for rejection of all units processed since the last monitored approval of the process.

4.3.6 *Inspection equipment.* The examination and tests shall be made using equipment listed on IEL 7548413 except as provided by MIL-I-45208.

4.4 Test methods and procedures.

4.4.1 *Mercurous nitrate.* The cartridge case, fitted with a steel mouth plug in accordance with fig. 3, shall be tested in accordance with the procedures prescribed by Standard Federal Test Method Standard No. 151. The cartridge case may be pickled for 30 seconds in a forty percent solution (specific gravity 1.25) of commercial nitric acid, in lieu of the sulphuric acid solution as specified.

4.4.2 *Hardness.* The cartridge case shall be sectionalized through the center of the head, three or more hardness readings shall be taken equally spaced through the head on both sides of the primer hole with the centerline of the hardness indents located 0.100 inches from the diameter of the primer hole. The hardness readings shall be made in accordance with the procedures prescribed by Standard Federal Test Method Standard No. 151.

4.4.3 *Tensile strength and elongation.* The test specimens shall be taken from the cartridge case in accordance with the standard position and size of tensile specimens chart, fig. 1. The tensile strength and elongation shall be determined in accordance with the procedures for these tests prescribed by Standard Federal Test Methods Standard No. 151.

4.4.4 *Macro examination.* The cartridge case shall be sectionalized through the center of the head, polished and examined for folds and cold shuts in the radius area of the head, in accordance with the procedures for this test prescribed by Standard Federal Test Method Standard No. 151. One hundred (100X) minimum magnification shall be used for this examination.

4.4.5 *Functional.*

4.4.5.1 *Initial production sample.*

4.4.5.1.1 *Excess pressure phase.* The cartridge case will be assembled into a complete round in the same manner as required for service firing. The propellant charge will be such as to produce the chamber pressure specified on the applicable drawing. The complete round will be temperature conditioned at plus 65° to plus 75° F. and fired in a serviceable weapon for which the round is standard. The weapon will be operated in the service manner.

4.4.5.1.2 *Excess, excess pressure phase.* The cartridge case will be assembled into a complete round in the same manner as required for service firing. The propellant charge will be such as to produce the maximum chamber pressure permitted by the proving ground. The complete round will be temperature conditioned at plus 65° to plus 75° F. and fired in a serviceable weapon for which the round is standard. The weapon will be operated in the service manner.

4.4.5.2 *Production sample.*

4.4.5.2.1 *Excess pressure phase.* The cartridge case will be assembled into a complete round in the same manner as required for service firing. The propellant charge will be such as to produce the chamber pressure specified on the applicable drawing. The complete round will be temperature conditioned at plus 65° to plus 75° F. and fired in a serviceable weapon for which the round is standard. The weapon will be operated in the service manner.

5. PREPARATION FOR DELIVERY

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5.1 Packing.

5.1.1 *Level A.* Not applicable.

5.1.2 *Level B.* Not applicable.

5.1.3 *Level C.* Cartridge cases shall be packed in accordance with requirements of drawing F-7543320.

5.2 **Labeling and marking.** Labeling and marking for shipment shall be in accordance with MIL–P–10025.

5.3 **Data card.** Data card information shall be as specified in MIL–A–2550.

6. NOTES

6.1 Prior to fabrication of an initial production sample the contractor upon approval of the contracting officer, may make and submit for evaluation up to 5 process control lots. These process control lots shall consist of not less than 10 nor more than 25 cartridge cases each. The contractor shall furnish all information pertaining to the manufacture of the lots to the contracting officer. All costs for [illegible] process control lots will be borne by the Government.

6.2 It is not implied that cartridge cases that contain minimum strength values for all specimens will function satisfactorily in ballistic test.

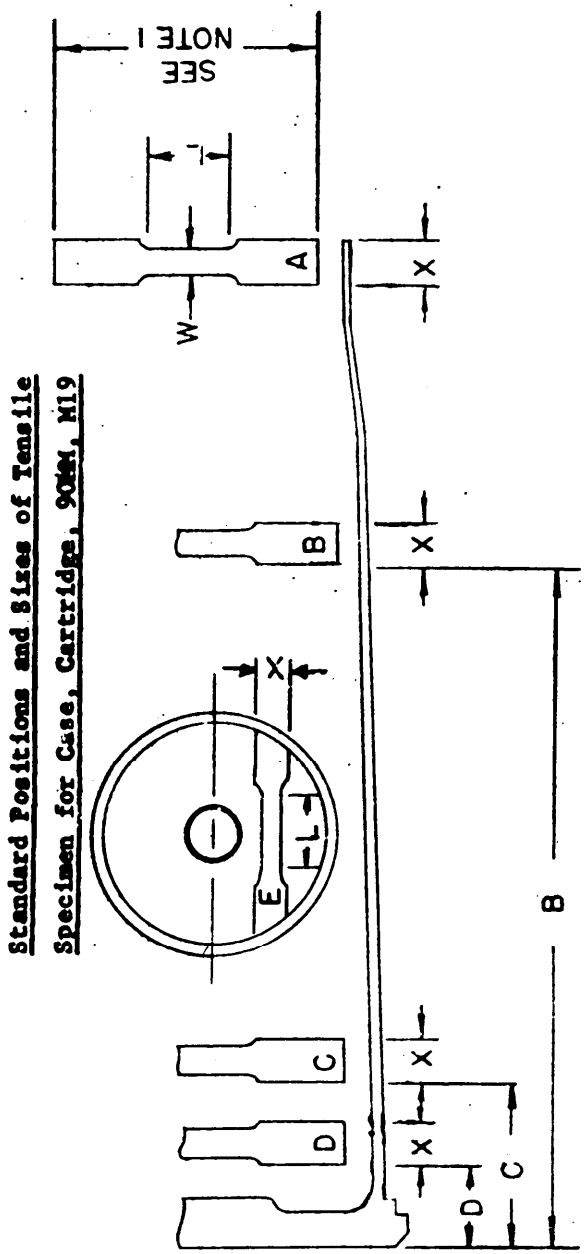
6.3 The term "ejection" shall mean that the fired cartridge case is thrown free of the breech of the weapon to permit loading of the subsequent round.

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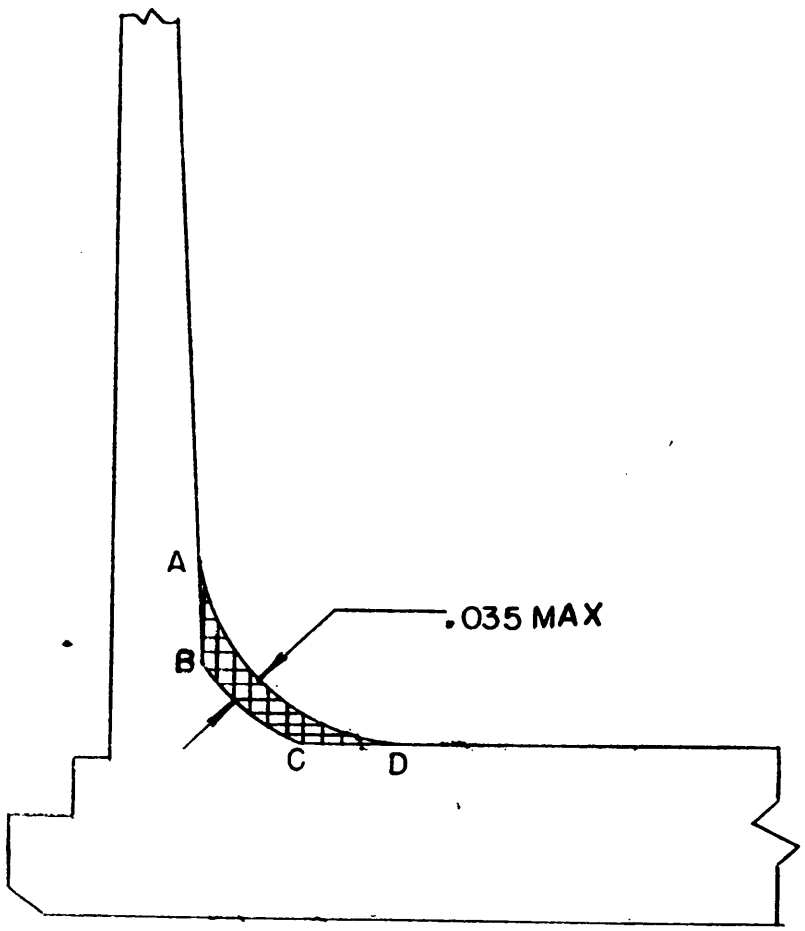
Standard Positions and Sizes of Tensile Specimen for Case, Cartridge, 90mm, M19

Notes:
1. Overall length of specimen to suit fixture in testing machine
2. Specimen E to be machined so as to eliminate effects of stamping on test
3. Specimen C and D to be machined to uniform thickness
4. Specimen E to be taken midway between primer hole counterbore and edge of flange.

Specimen	X	W	L	Location From Base
A	1.25	1.0	2.0	At Mouth
B	1.25	1.0	2.0	15.25
C	1.25	1.0	2.0	2.00
D	1.25	1.0	2.0	0.57
E	0.40	0.3	1.0	In Base

FIGURE 1.

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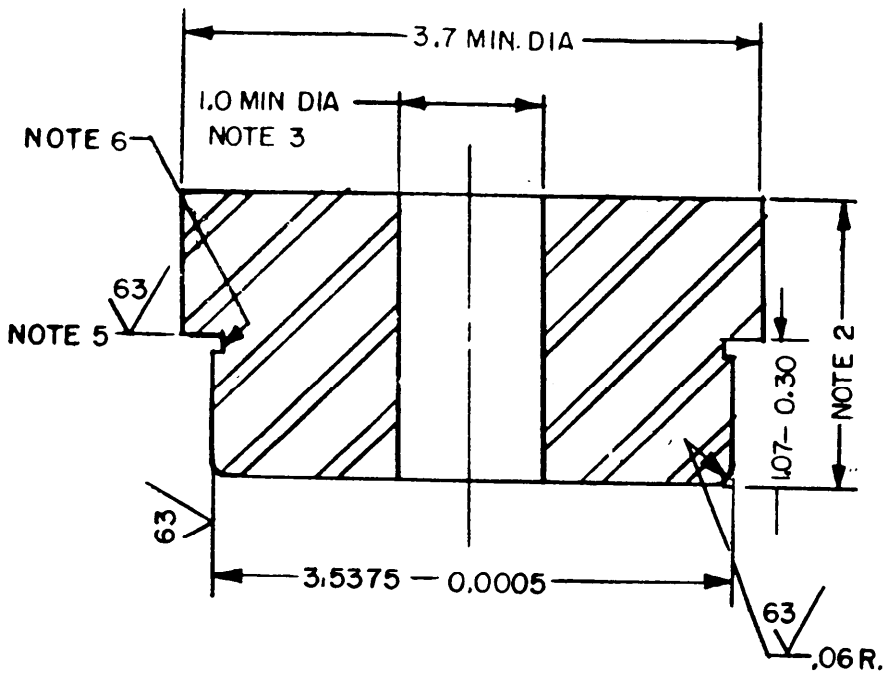


MACRO EXAMINATION

Folds or Cold Shuts shall not extend beyond shaded area inclosed by tangent AB, tangent CD and Arc BC.

FIGURE 2.

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NOTES:

1 - Material
2 - Overall length to suit testing procedure
3 - Hole may be threaded for extracting plug
4 - 250 finish except as noted
5 - Cartridge case to be seated against this surface
6 - 1/16 X 1/16 inch undercut

MOUTH PLUG FOR USE IN MERCUROUS NITRATE TEST

FIGURE 3.

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